

→ visible units activate the hidden unit.

Applications:

* Feature learning, Recommendation systems.

Autoencoders:

* An autoencoder is a neural network that learns to reproduce its input.

→ It has three main parts:

Input → Encoder → Bottle neck → Decoder

→ Reconstructed output

Working:

(i) The encoder convert the input into a smaller representation.

(ii) The decoder reconstructs the original input.

(iii) The network is trained to minimize this loss.

Unsupervised Learning, RBMs and Autoencoders.

Unsupervised Learning:

Unsupervised learning is a type of machine learning where the model learns patterns or structures from unlabelled data.

Example:

Clustering, Dimensionality reduction, Feature learning, Anomaly detection.

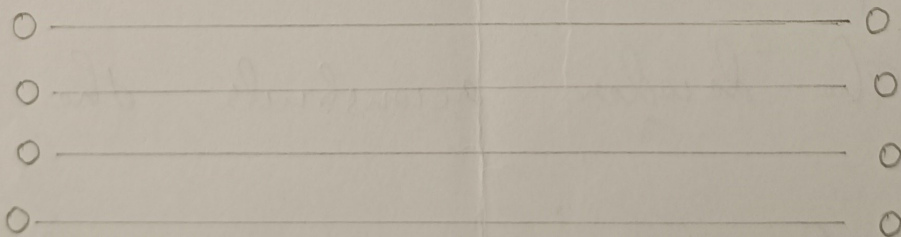
Restricted Boltzmann Machine (RBM):

An RBM is a generative neural network that learns the probability distribution of input data.

It contains two layers:

Visible layer

Hidden layer



Visible Layer: Represent the input data.

Hidden Layer: learns important features.

Working Principle:

→ Input data is given to the visible layer.

Leaky ReLU (LReLU): Leaky ReLU allows a small negative output instead of making it completely zero.

Formula:

$$f(x) = \begin{cases} x, & x \geq 0 \\ \alpha x, & x < 0 \end{cases}$$

Advantage

- Reduces the dying ReLU problem
- Easy to implement

Disadvantages.

- Allow a small gradient for negative inputs.
- May not always perform better than ReLU.

ELU - Exponential Linear Unit:

Formula →

$$f(x) = \begin{cases} x & x \geq 0 \\ \alpha(e^x - 1), & x < 0 \end{cases}$$

Advantage.

- Provides non-zero gradient for negative inputs.

- Requires an additional parameter α .

Disadvantage.

- More computationally expensive than ReLU because it uses an exponential function.
- Reduces the dying neuron problem.

Compare ReLU, Leaky ReLU and ELU:

Activation functions introduce non-linearity into neural network and help them learn complex patterns.

1. ReLU - Rectified Linear Unit:

Formula $\rightarrow$ $f(x) = \max(0, x)$.

Therefore;

if $x > 0$, output $= x$.

if $x \leq 0$, output $= 0$.

Advantage:

- $\rightarrow$ Simple and computationally efficient.
- $\rightarrow$ Helps reduce the vanishing-gradient problem for positive inputs.
- $\rightarrow$ Widely used in deep neural networks.

Disadvantage:

- $\rightarrow$ For negative inputs the gradient becomes zero. This can cause the dying ReLU problem, where some neurons stop learning.

simultaneously.

Advantages.

- Can represent more complex patterns.
- Provides greater learning capacity.

Disadvantages.

- Increase the number of Parameters.
- Requires more memory and computation.

Depth of neural networks:

Depth refers to the number of layers in a neural network.

Example:

Input $\rightarrow$ Hidden $\rightarrow$ Hidden 2 $\rightarrow$ Hidden 3

$\rightarrow$ Output.

Effect on learning capability and performance:

$\rightarrow$ Increasing width or depth generally increases computational complexity and training time.

$\rightarrow$ Too large a network may overfit the training data.

$\rightarrow$ A well-designed deep network can achieve better performance on complex task.

Representation Learning width and Depth of Neural Networks.

Representation Learning:

Representation learning is a technique in which a neural network automatically learns useful features or representations from raw data instead of requiring humans to 'manually' define the features.

For example in image recognition:

Raw Image $\rightarrow$ Edges $\rightarrow$ Shapes $\rightarrow$ Objects $\rightarrow$ class.

* A deep neural network can automatically learn these levels of representation.

Width of a neural network:

* Width refers to the number of neurons in a layer.

Example:

Input $\rightarrow$ ○ ○ ○ ○ ○

Hidden layer

○ ○ ○ ○ ○ ○ ○

Deep Learning:

Deep Learning.

Features

Architecture

- Uses multi layer neural networks with many hidden layers.

Data Requirement

- Usually requires large amount of Data.

Feature Extraction

- Features are automatically learned from raw data.

Training Time

- Usually requires more training time

Computational Power

- Usually benefits from GPUs/TPUs.

Performance

- Excellent for complex data such as Images, audio and text.

Interpretability

- Usually more difficult to interpret.

Machine Learning applications:

✓ Email Spam detection - classifier email as spam or genuine.

Deep learning applications:

✓ Face recognition - Identifier people from Images or videos.

1) Differentiate Between Machine Learning and Deep Learning.

Machine Learning (ML) and Deep Learning (DL) are methods of Artificial Intelligence used to learn patterns from data.

Machine Learning:

Features

Machine Learning

Architecture

- Uses algorithms such as Decision Trees, SVM, KNN and shallow neural networks.

Data Requirement

- Can work with relatively smaller datasets.

Feature Extraction

- Features are generally selected/extracted manually.

Training Time

- Generally faster.

Computational Power

- Can offer run on CPUs.

Performance

- Good for structured / tabular data.

Interpretability

- Often easier to interpret.

Assessment Tool -> 03

CO2 - AT3 - Class Test.

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